Class 9 Mathematics — Exercise 3.3

Basic Concept

(1) Ordered Pair

A pair of elements written in a particular **order** inside small brackets, e.g. (a, b). Here a is the **first element** and b is the **second element**.

$$(2, 5) \neq (5, 2)$$

Equality of ordered pairs: $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

(2) Cartesian Product

$$A \times B = \{(x, y) \mid x \in A \wedge y \in B\}$$

- In $A \times B$, first elements come from A and second elements come from B.
- $A \times B \neq B \times A$ (in general).
- $n(A \times B) = n(A) \times n(B)$
- $A \times \phi = \phi, \quad \phi \times A = \phi$

Question 1

Find the values of unknowns when:

(i) $(a, -b) = (7, 1)$

Equating corresponding elements:

$$a = 7$$

$$-b = 1 \Rightarrow b = -1$$

$$\boxed{a = 7, \quad b = -1}$$

(ii) $(2a, 2b + 3) = (-10, -b)$

Equating corresponding elements:

$$2a = -10 \Rightarrow a = -5$$

$$2b + 3 = -b \Rightarrow 2b + b = -3 \Rightarrow 3b = -3 \Rightarrow b = -1$$

$$\boxed{a = -5, \quad b = -1}$$

(iii) $(2a - 4, 6) = (8, -b + 1)$

Equating corresponding elements:

$$2a - 4 = 8 \Rightarrow 2a = 12 \Rightarrow a = 6$$
$$6 = -b + 1 \Rightarrow b = 1 - 6 \Rightarrow b = -5$$

$$\boxed{a = 6, \quad b = -5}$$

(iv) $(x + 2y, y - 3) = (2, 5)$
Equating corresponding elements:

$$y - 3 = 5 \Rightarrow y = 8$$
$$x + 2y = 2 \Rightarrow x + 16 = 2 \Rightarrow x = -14$$

$$\boxed{x = -14, \quad y = 8}$$

(v) $(2x - y, y - 3x) = (4, 2)$
Equating corresponding elements:

$$2x - y = 4 \quad \ldots(1)$$
$$y - 3x = 2 \Rightarrow y = 3x + 2 \quad \ldots(2)$$
$$\text{Put (2) in (1): } 2x - (3x + 2) = 4 \Rightarrow -x - 2 = 4 \Rightarrow x = -6$$
$$y = 3(-6) + 2 = -16$$

$$\boxed{x = -6, \quad y = -16}$$

(vi) $(4x + 6y, x - 12y) = (6, -3)$
Equating corresponding elements:

$$4x + 6y = 6 \quad \ldots(1)$$
$$x - 12y = -3 \Rightarrow x = 12y - 3 \quad \ldots(2)$$
$$\text{Put (2) in (1): } 4(12y - 3) + 6y = 6 \Rightarrow 48y - 12 + 6y = 6 \Rightarrow 54y = 18 \Rightarrow y = \frac{1}{3}$$
$$x = 12\left(\frac{1}{3}\right) - 3 = 4 - 3 = 1$$

$$\boxed{x = 1, \quad y = \frac{1}{3}}$$

(vii) $(5x + y, -x + y) = (6, 1)$
Equating corresponding elements:

$$5x + y = 6 \quad \ldots(1)$$
$$-x + y = 1 \quad \ldots(2)$$

$$(1) - (2): \; 6x = 5 \Rightarrow x = \frac{5}{6}$$

$$\text{From } (2): \; y = 1 + x = 1 + \frac{5}{6} = \frac{11}{6}$$

$$\boxed{x = \frac{5}{6}, \quad y = \frac{11}{6}}$$

Question 2

Let $A = \{1, 4, 8\}$ and $B = \{1, 0\}$. Find (i) $A \times B$ (ii) $B \times A$ (iii) $A \times A$ (iv) $B \times B$. How many elements are there in $A \times B$, $B \times A$, $A \times A$ and $B \times B$?

(i) $A \times B = \{(1,0), (1,1), (4,0), (4,1), (8,0), (8,1)\}$

$n(A \times B) = n(A) \times n(B) = 3 \times 2= \boxed{6}$ elements

(ii) $B \times A = \{(0,1), (0,4), (0,8), (1,1), (1,4), (1,8)\}$

$n(B \times A) = n(B) \times n(A) = 2 \times 3 = \boxed{6}$ elements

(iii) $A \times A = \{(1,1), (1,4), (1,8), (4,1), (4,4), (4,8), (8,1), (8,4), (8,8)\}$

$n(A \times A) = n(A) \times n(A) = 3 \times 3 = \boxed{9}$ elements

(iv) $B \times B = \{(0,0), (0,1), (1,0), (1,1)\}$

$n(B \times B) = n(B) \times n(B) = 2 \times 2 = \boxed{4}$ elements

Question 3

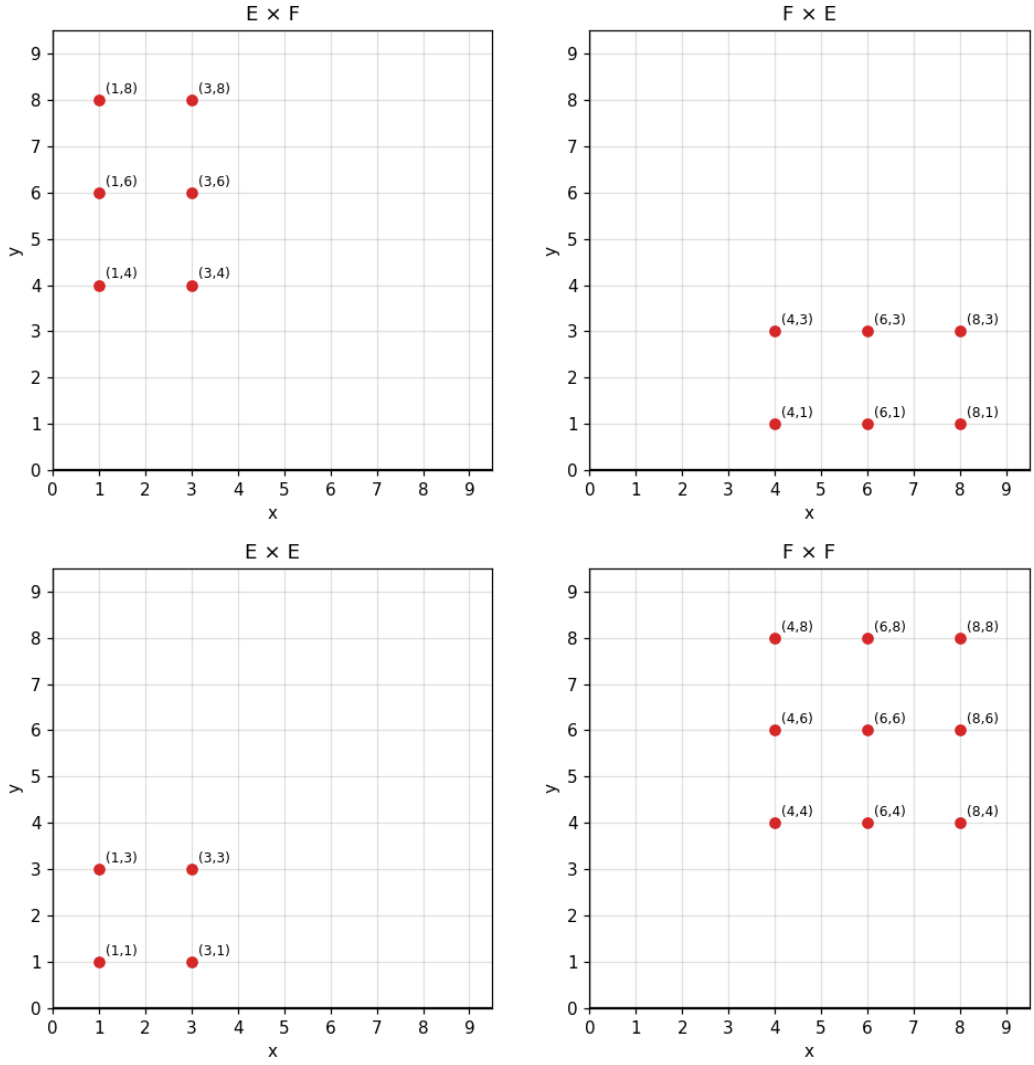
Let $E = \{1, 3\}$ and $F = \{4, 6, 8\}$. Express $E \times F$, $F \times E$, $E \times E$, $F \times F$ graphically.

$$E \times F = \{(1,4), (1,6), (1,8), (3,4), (3,6), (3,8)\}$$

$$F \times E = \{(4,1), (4,3), (6,1), (6,3), (8,1), (8,3)\}$$

$$E \times E = \{(1,1), (1,3), (3,1), (3,3)\}$$

$$F \times F = \{(4,4), (4,6), (4,8), (6,4), (6,6), (6,8), (8,4), (8,6), (8,8)\}$$



Question 4

If $L \times M = \{(0,2),(0,3),(0,4),(1,2),(1,3),(1,4)\}$, then find sets L, M and $M \times L$.

L = set of all first elements, M = set of all second elements

$$L = \{0,1\}, \qquad M = \{2,3,4\}$$

$$M \times L = \{(2,0),(2,1),(3,0),(3,1),(4,0),(4,1)\}$$

Question 5

Given that $A = \{1,3,5\}$, $B = \{2,4\}$, $C = \{6,7\}$. (i) Find $A \times (B \cup C)$ (ii) Find $(A \times B) \cup (A \times C)$ (iii) Verify $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(i) $B \cup C = \{2,4,6,7\}$

$$A \times (B \cup C) = \{(1,2),(1,4),(1,6),(1,7),(3,2),(3,4),(3,6),(3,7),(5,2),(5,4),(5,6),(5,7)\}$$

(ii) $A \times B = \{(1,2),(1,4),(3,2),(3,4),(5,2),(5,4)\}$

$A \times C = \{(1,6),(1,7),(3,6),(3,7),(5,6),(5,7)\}$

$$(A\times B)\cup(A\times C) = \{(1,2),(1,4),(1,6),(1,7),(3,2),(3,4),(3,6),(3,7),(5,2),(5,4),(5,6),(5,7)\}$$

(iii) From (i) and (ii), both sides have the same 12 ordered pairs, so

$$\boxed{A \times (B \cup C) = (A \times B) \cup (A \times C)}$$

Question 6

Given that $D = \{a, e, i\}$, $E = \{a, c\}$, $F = \{b, c\}$. **(i) Find** $D \times (E \cap F)$ **(ii) Find** $(D \times E) \cap (D \times F)$ **(iii) Verify** $D \times (E \cap F) = (D \times E) \cap (D \times F)$

(i) $E \cap F = \{c\}$

$$D \times (E \cap F) = \{(a, c), (e, c), (i, c)\}$$

(ii) $D \times E = \{(a, a), (a, c), (e, a), (e, c), (i, a), (i, c)\}$

$D \times F = \{(a, b), (a, c), (e, b), (e, c), (i, b), (i, c)\}$

$$(D \times E) \cap (D \times F) = \{(a, c), (e, c), (i, c)\}$$

(iii) From (i) and (ii):

$$\boxed{D \times (E \cap F) = (D \times E) \cap (D \times F)}$$

Question 7

Given that $A = \{x \mid x \in N, x < 3\}$, $B = \{y \mid y \in W, y < 2\}$, $C = \{0, 2, 4\}$. **(i) Verify** $A \times (B - C) = (A \times B) - (A \times C)$ **(ii) Verify** $(A - B) \times C = (A \times C) - (B \times C)$

$$A = \{1, 2\}, \qquad B = \{0, 1\}, \qquad C = \{0, 2, 4\}$$

(i) L.H.S: $B - C = \{1\}$

$$A \times (B - C) = \{(1, 1), (2, 1)\}$$

R.H.S: $A \times B = \{(1, 0), (1, 1), (2, 0), (2, 1)\}$

$A \times C = \{(1, 0), (1, 2), (1, 4), (2, 0), (2, 2), (2, 4)\}$

$$(A \times B) - (A \times C) = \{(1, 1), (2, 1)\}$$

L.H.S = R.H.S, hence verified.

(ii) L.H.S: $A - B = \{2\}$

$$(A - B) \times C = \{(2, 0), (2, 2), (2, 4)\}$$

R.H.S: $A \times C = \{(1, 0), (1, 2), (1, 4), (2, 0), (2, 2), (2, 4)\}$

$B \times C = \{(0, 0), (0, 2), (0, 4), (1, 0), (1, 2), (1, 4)\}$

$$(A \times C) - (B \times C) = \{(2, 0), (2, 2), (2, 4)\}$$

L.H.S = R.H.S, hence verified.

Question 8

Let $X = \{x \mid x \in W, x \leq 2\}$ and $Y = \{-1, -2, -3\}$. Exhibit $X \times Y$ and $Y \times X$ by arrow diagram.

$$X = \{0, 1, 2\}, \qquad Y = \{-1, -2, -3\}$$

$$X \times Y = \{(0, -1), (0, -2), (0, -3), (1, -1), (1, -2), (1, -3), (2, -1), (2, -2), (2, -3)\}$$

$$Y \times X = \{(-1, 0), (-1, 1), (-1, 2), (-2, 0), (-2, 1), (-2, 2), (-3, 0), (-3, 1), (-3, 2)\}$$

Arrow diagrams: (each arrow goes from the first element to the second element)

